

navigation, organised around libraries, cells and cell views.

Other important features are the ability to create parametric cells, support for global nets, buses and bundles, enhanced metafile format schematic output for enhanced printing functionality and options for autowiring and on-canvas editing of element properties.

The free edition of Symica has almost the same functionalities as the full version, with just a limit to the number of devices that can be simulated and lack of support for mixed signals.

The latest beta version release of Symica also has certain added features.

Component description format helps the user specify the component occurrence in a netlist and allows him/her to create different netlists for different formats such as SPICE and LVS.

The feature, probing currents for subcircuit output pins, allows probing currents for complex transistor models that are represented as subcircuits. Two modes of probing were realised, direct plot was elaborated and hierarchical nets highlights have been added to this version.

Another useful feature introduced was the ability to share a project amongst multiple users. This is particularly favourable amongst design teams spread across different geographies. One can also choose the views that would be made available to each user.

Easy to use yet packed with guides

The user interface of Symica is simple and intuitive, and similar to that of the IC design toolkits provided by the top EDA tools for IC design and simulation. The Symica package comes with extensive help files, which are automatically made available with software installation.

There are six help guides available with installation, one for each of the six products (Symica DE, SymSpice, SymProbe, Symbol Wizard, SKILL Translator and SymLayout) in the Symica toolkit. Help files can be accessed from the toolbar, Help > Symica Help.

What does it come with?

Symica DE: Symica Design Environment is a platform for circuit design. It integrates all necessary tools for simulation. It also facilitates the preparation and execution of simulations and the inspection and interpretation of the results of simulation.

SymSpice: This is an original SPICE simulation program developed by Symica. It offers 'golden' SPICE level of simulation and accuracy. It is compatible with commonly used industry netlist formats and supports almost all commonly used transistor models.

SymSpice Turbo: This is Symica's fast-SPICE simulator. It is specifically designed to work with very large-scale integrated circuits and systems on chips. Symica's proprietary math methods provide high accuracy and considerably good speed. The time reduction factor is highly dependent on the type of design.

SymProbe: It is the waveform viewer and analogue simulation result analyser. It comes with all the functions expected in a convenient graphical user interface.

Symbol Wizard: This is a process design kit (PDK) generator that helps you create libraries with symbols of the primitive devices (such as MOS and bipolar transistors, diodes and resistors) in clear and efficient format for your particular technology process. This can be created in Symica format from spice model decks either in SPICE or Spectre format.

SKILL Translator: This is an import and export tool that lets translate libraries in Cadence to Symica and vice versa. Users can load either the entire design or parts of a design project previously developed in different formats to Symica platform.

From the dropdown menu, select the help guide you want to view. These PDFs give a detailed explanation of their respective functions and procedures, hence making the software simple even for users with only basic knowledge of SPICE.

Build your first system on chip

The main fields of application for Symica tools are chip design, verification

and testing of large circuits and systems on chips (SoCs), and post layout circuit simulation. The chip design includes multi-bit and high-speed analogue circuits that require long test sequences, projects with mixed signal analogue-to-digital conversion, large digital circuits that require analogue simulation for reaching peak performance and all kinds of analogue IPs.

Symica has a silicon-proven technology. Hundreds of chips are already designed and manufactured using Symica tools, including analogue-to-digital (ADC) and digital-to-analogue converters (DAC), phase locked loops (PLL) and high-speed controllers. Numerous circuits have been simulated with SymSpice with around two thou-

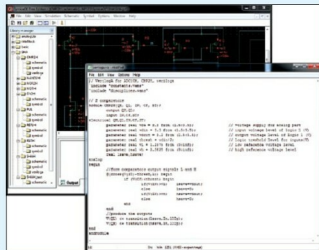


Fig. 4: Symica supports Verilog-A models

sand of these being used for regression testing on each new SymSpice version.

Competing tools from the big three

Symica is a Ukraine-based electronic design automation (EDA) company that was founded in 2009. The Symica DE provided by this startup has competitors that include the big three in the EDA industry—Synopsys, Cadence and Mentor Graphics.

HSPIICE by Synopsys, Spectre Circuit Simulator by Cadence, Pyxis Custom IC Solution of Mentor Graphics, Analogue/Mixed-Signal/RF and Custom IC CAD by Silvaco, Quite Universal Circuit Simulator (Qucs) and Ngspice are a few other EDA tools